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Inventor(s)

Herron; Brian J. et al.

Vehicle diagnostic system and method with vehicle system identification and checklist generation

Abstract

A vehicle diagnostic system and method for identifying and presenting vehicle safety systems on a vehicle for calibration by an operator includes providing a vehicle diagnostic computer tool that includes a display screen and is configured to be connected to a diagnostic port of a vehicle. The vehicle diagnostic tool is configured to determine vehicle safety systems present on the vehicle and to receive vehicle damage information inputs corresponding to areas on the vehicle in which there is damage. A system list program generates a list of vehicle safety systems equipped on the vehicle requiring calibration based on the determined vehicle safety systems present on the vehicle and the vehicle damage information inputs, where the list corresponds to vehicle safety systems present on the vehicle that are disposed in damage areas. The vehicle diagnostic computer tool is further configured to display the list on the screen.

Inventors: Herron; Brian J. (Dexter, MI), Horak; Jay H. (Mesa, AZ), Jurmo; Michael D. (Canton, MI), Joseph; David M. (Ypsilanti, MI), Devericks; Michael B. (Ventura, CA)

Applicant: Opus IVS, Inc. (Ann Arbor, MI)

Family ID: 96950460

Assignee: Opus IVS, Inc. (Dexter, MI)

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Primary Examiner: Mirza; Adnan M

Attorney, Agent or Firm: Gardner, Linn, Burkhardt & Ondersma LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) The present application is a continuation-in-part of U.S. application Ser. No. 17/509,316, filed Oct. 25, 2021, and is a continuation-in-part of U.S. application Ser. No. 17/515,516, filed on Oct. 31, 2021, both of which are hereby incorporated herein by reference in their entireties.

BACKGROUND OF THE INVENTION

(1) The present invention is directed to a vehicle diagnostic system and method, and in particular to a vehicle diagnostic system that identifies electronic systems equipped on the vehicle and generates a listing of systems that may require repair or calibration based on vehicle damage and configuration, as well as identifies calibration functions supported by the vehicle diagnostic system.

(2) Vehicle diagnostic systems employing diagnostic scan devices or tools are used in automotive repair facilities to diagnose and repair computer-based vehicle systems, where vehicles may have differing computer-based systems depending on the configuration and options installed on the vehicle. Vehicle diagnostic scan systems may include or use one or more diagnostic software scanning programs or applications, such as applications developed by an OEM or an aftermarket diagnostic company. OEM diagnostic applications may include proprietary diagnostic application software exclusive to a particular OEM, and may be required for certification purposes to validate repairs performed on a vehicle.

SUMMARY OF THE INVENTION

(3) The present invention provides a vehicle diagnostic system and method, and in particular a diagnostic system that utilizes a vehicle diagnostic computer tool to identify electronic systems equipped on a vehicle and generate a list of systems that may require repair or calibration based on the configuration of the vehicle and any damage to the vehicle, as well as identifies calibration functions supported by the vehicle diagnostic system.

(4) A method for identifying and presenting vehicle safety systems on a vehicle for calibration by an operator in accordance with an aspect of the present invention comprises providing a vehicle diagnostic computer tool that includes a display screen and is configured to be connected to a diagnostic port of a vehicle to be in communication with an electronic system of the vehicle, as well as includes providing a system list program that is operable to generate a list of vehicle safety systems equipped on the vehicle requiring calibration based on inputs to the system list program. The method further includes determining vehicle safety systems present on the vehicle via the vehicle diagnostic computer tool, determining areas on the vehicle in which there is damage, and providing vehicle damage information inputs to the system list program based on the damage areas. The method additionally includes generating a list of vehicle safety systems equipped on the vehicle requiring calibration via the system list program based on the determined vehicle safety systems present on the vehicle and the vehicle damage information inputs, where the list of vehicle safety systems equipped on the vehicle requiring calibration corresponds to vehicle safety systems present on the vehicle that are disposed in damage areas, and the method further includes providing the list of vehicle safety systems equipped on the vehicle requiring calibration to a user. In one embodiment, for example, the list of vehicle safety systems equipped on the vehicle requiring calibration is provided by being displayed on a screen of the vehicle diagnostic computer tool. Alternatively, or in addition, the list of vehicle safety system equipped on the vehicle requiring calibration may be output to a report. For example, an electronic data file may be generated by the vehicle diagnostic computer tool that may be provided to another computer device and/or may be printed.

(5) In particular embodiments, the method includes reading vehicle data information with the vehicle diagnostic computer tool and determining the vehicle safety systems present on the vehicle using the vehicle data information, where the vehicle safety systems determined to be on the vehicle may be those present on the vehicle as built or as equipped by the vehicle manufacturer. The determination of the vehicle safety systems present on the vehicle may be done via an onboard configuration determination or an offboard configuration determination, such as via one or more onboard or offboard databases. The onboard configuration determination may be obtained from the vehicle itself or via a database located on the vehicle diagnostic computer tool, and the offboard configuration determination may be performed by accessing a separate database identifying the as built configuration of the vehicle, such as by way of a vehicle identification number (VIN) of the vehicle. As such, the vehicle data information may comprise a VIN of the vehicle, and an offboard database may be used to identify vehicle systems present on the vehicle, such as by way of the VIN.

(6) In accordance with a further aspect of the invention, the vehicle damage information inputs may comprise at least one of (i) one or more digital images of the vehicle, where image recognition software is used on the one or more digital images to determine damage to the vehicle, (ii) data from a collision estimating software, and/or (iii) data input by a user of the vehicle diagnostic tool.

(7) Still further, the vehicle diagnostic system is configured to support calibration of one or more of the listed vehicle safety systems requiring calibration. In a particular configuration, the system displays an indication of which of the vehicle safety systems requiring calibration are supported by the vehicle diagnostic system. The vehicle diagnostic computer tool is further configured to enable a user to access and launch the calibration support information with the vehicle diagnostic computer tool for calibration of vehicle safety systems requiring calibration that are supported by

the vehicle diagnostic system. The calibration support information may comprise human readable instructions contained within electronic data files stored on the vehicle diagnostic computer tool that may be displayed or printed, and may include calibration programs, such as executable programs for interacting with the electronic system of the vehicle.

(8) The vehicle diagnostic computer tool may further be configured to require confirmation that all of the listed vehicle safety systems equipped on the vehicle requiring calibration have been calibrated prior to running a vehicle diagnostic scan with the vehicle diagnostic computer tool, such as prior to running a post scan.

(9) In accordance with a further aspect of the present invention, a vehicle diagnostic system comprises a vehicle diagnostic computer tool that includes a display screen and is configured to be connected to a diagnostic port of a vehicle to be in communication with an electronic system of the vehicle, and includes a system list program. The vehicle diagnostic tool is configured to determine vehicle safety systems present on the vehicle and is configured to receive vehicle damage information inputs corresponding to areas on the vehicle in which there is damage. The system list program is configured to generate a list of vehicle safety systems equipped on the vehicle requiring calibration based on the determined vehicle safety systems present on the vehicle and the vehicle damage information inputs, where the list of vehicle safety systems equipped on the vehicle requiring calibration corresponds to vehicle safety systems present on the vehicle that are disposed in damage areas. The vehicle diagnostic computer tool is further configured to display on the screen the list of vehicle safety systems equipped on the vehicle requiring calibration.

(10) The vehicle diagnostic system and method of the present invention provides a checklist to a mechanic repairing a damaged vehicle regarding ADAS safety systems that may require calibration, as well as enables the mechanic to launch calibration functions using the vehicle diagnostic computer tool to calibrate such systems that are supported by the vehicle diagnostic system. In this way, all of the ADAS systems requiring calibration may be tracked, with calibration thereof being confirmed prior to running a post scan operation. These and other objects, advantages, purposes and features of this invention will become apparent upon review of the following specification in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram of a vehicle diagnostic system in accordance with the present invention showing a vehicle diagnostic computer tool connected to the electronic system of a vehicle via a vehicle interface device;

(2) FIG. 2 is a block diagram of the vehicle diagnostic system in accordance with the present invention in relation to the electronic system of the vehicle;

(3) FIG. 3 is a block diagram of programs stored in memory, including a system checklist program, diagnostic evaluation program, and diagnostic scanning program applications of the vehicle diagnostic system of FIG. 1;

(4) FIG. 3A discloses an exemplary database correlating vehicle identification number data to vehicle systems for a vehicle;

(5) FIG. 3B discloses an exemplary database correlating vehicle identification number data to vehicle systems for a vehicle;

(6) FIG. 3C discloses exemplary calibration files for calibration of vehicle ADAS systems;

(7) FIG. 4 illustrates a vehicle system checklist displayed to a mechanic on a computer device that is generated by the vehicle diagnostic system in accordance with aspects of the present invention; and

(8) FIG. 5 is a flow chart illustrating aspects of the vehicle diagnostic method in accordance with

the present invention by which the vehicle diagnostic computer tool provides a system checklist to an operator.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

(9) The present invention will now be described with reference to the accompanying figures, wherein the numbered elements in the following written description correspond to like-numbered elements in the figures.

(10) A vehicle diagnostic system **10** for use with a vehicle **22** is shown for use by a user such as a mechanic or operator **24**, such as in an automotive repair facility. System **10** is illustrated in FIGS. **1** and **2** as including a vehicle diagnostic computer tool **28**, which as discussed in more detail below, is connected with vehicle **22** by operator **24** in order to diagnose the electronic system **23** of vehicle **22**, including various vehicle electronic control units (ECUs), including safety systems such as Adaptive Driver Assistance Systems (“ADAS”) ECUs, and including other electronic parts and components of vehicle **22**. Vehicle diagnostic tool **28** may be used to scan the electronic system **23** of vehicle **22**, such as to determine any fault codes in the electronic system **23**, which may be reported as diagnostic trouble codes (“DTCs”). A vehicle scan may be run prior to repairs, such as part of an initial diagnosis to provide an indication to the mechanic **24** as to what repairs are needed on vehicle **22**, and/or may be run after performing repairs, such as to confirm that vehicle **22** has been properly repaired.

(11) As discussed in more detail below, diagnostic computer tool **28** is configured to determine the electronic vehicle systems with ECUs equipped on vehicle **22**, such as in particular safety systems, including ADAS systems, and is further configured to receive inputs **35** corresponding to damage to vehicle **22**, such as from a collision, with the diagnostic computer tool **28** in turn generating a vehicle system list or system checklist **26** (FIG. **4**) identifying and presenting to the mechanic **24** electronic vehicle systems of vehicle **22** that may need calibration, where the status of completion of the indicated calibrations can be readily tracked and recorded to confirm completion of repairs. According to still additional features, diagnostic computer tool **28** may support calibration functions of various electronic vehicle systems of vehicle **22**, with the system checklist **26** in turn including an identification for the mechanic **24** of the supported calibration functions from which the mechanic **24** can select and launch using diagnostic computer tool **28**.

(12) As discussed in more detail below, diagnostic system **10** utilizing diagnostic computer tool **28** is thus operable based on inputs to diagnostic tool **28** to provide a checklist **26** to mechanic **24** of systems on vehicle **22** with ECUs, such as safety systems including ADAS systems and airbag modules, that may require calibration based on the determined configuration of vehicle **22** and damage thereto, with diagnostic tool **28** additionally identifying for mechanic **24** any such systems for which diagnostic tool **28** supports calibration for that vehicle **22**. The checklist **26** is generated by a system list or system checklist program **45** that may additionally be configured to communicate with one or more remote computer systems, such as with a sever **64** via an Internet connection **70**, including for example to confirm completion of calibrations and to automatically toggle items on the checklist **26** as completed when calibrations have been performed. It should be appreciated, however, that diagnostic computer tool **28** may itself be able to confirm completion of calibrations and automatically toggle items on checklist **26**, such as for calibration operations supported on diagnostic computer tool **28**.

(13) Vehicle diagnostic computer tool **28** is configured in the illustrated embodiment as comprising a computer device **30**, such as a laptop or tablet computer that includes circuitry, hardware and software, and is coupled with a vehicle interface module **27** by a cable **42a**. In use, tool **28** is connected with vehicle **22** by operator **24**, such as by connecting to an on-board diagnostic (“OBD”) diagnostic port **32** of the vehicle **22** in order to diagnose the electronic system **23** of vehicle **22**, including the various noted vehicle systems comprising vehicle electronic control units (ECUs), such as an engine ECU **34**, body ECU **36**, brakes ECU **38**, and/or other ECUs **40**, including for safety systems such as ADAS ECUs **41**, and including other electronic parts and

components of vehicle **22**, including other safety systems such as airbag modules and the like. Tool **28** connects with port **32** via interface **27**, such as via vehicle cable **42b**. Vehicle interface **27** includes a controller **29**, such as in the form of a processor or micro-processor and interface circuitry to facilitate communication between the ECUs and tool **28**, with interface **27** including a database of vehicle protocols found in a local memory **44** that allow communication with the ECUs of various makes and models of vehicles. Vehicle interface **27** additionally includes an interface **46** for communication between interface module **27** and computer **30** via interface **74** of computer **30**, where computer **30** additionally includes a controller **48** and memory **50**. Diagnostic tool **28** additionally includes a user interface **66**, which may comprise a touch screen **66a** and/or keyboard **66b** of computer **30** enabling mechanic **24** to enter information into diagnostic tool **28**, as well as view information output by diagnostic tool **28**. As understood from FIGS. **1** and **2**, vehicle diagnostic tool **28** may be connected to a remote computer, such as a server **64**, such as by an Internet **70** connection. Although vehicle diagnostic computer tool **28** is shown as comprising a separate computer **30** coupled with vehicle interface device **27** in the illustrated embodiment, it should be appreciated that in an alternative arrangement computer **30** and vehicle interface device **27** may be integrated together, such as for example, in an arrangement such as disclosed in commonly assigned U.S. patent application Ser. No. 16/701,967, which is incorporated herein by reference in its entirety.

(14) Vehicle diagnostic system **10** may be configured to be selectively operable in one of a plurality of different modes, whereby a technician may use the diagnostic tool **28** for vehicle maintenance, diagnosis, programming and repair as needed. In the illustrated embodiment, and as understood from FIGS. **2** and **3** in accordance with an embodiment of the present invention, system **10** includes or provides access to various programs **50a** comprising diagnostic applications, such as stored in a memory **50** of diagnostic computer tool **28**, including system list program **45**, a diagnostic evaluation program application **51**, and commercially available diagnostic application scanning programs **53a**, **53b** that are configured for use with the specific vehicle **22** and may be used or selected by a mechanic **24**. In the illustrated embodiment diagnostic application program **53a** comprises an aftermarket scan program and diagnostic application program **53b** comprises an OE scan program. Although shown as having two diagnostic application programs **53a**, **53b**, it should be appreciated that system **10** may include more than two such programs available for use with a given vehicle **22** and/or including additional diagnostic application programs for use with other vehicles, depending on make, model and/or year of a given vehicle. Interface module **27** may be configured as an SAE standard J2534 device, such as a device compliant with the J2534-2 standard, or as an ISO compliant or other standard compliant device for supporting and enabling communication with the electronic systems of a vehicle.

(15) As illustrated in FIGS. **2** and **3**, the diagnostic application scanning programs **53a**, **53b** are stored in a memory **50**. An exemplary aftermarket diagnostic application scanning program **53a** comprises a program provided by an automotive manufacturer or a company that supplies diagnostic application programs, such as Opus IVS, Inc. of Ann Arbor, Michigan, and the OE diagnostic application scanning program **53b** comprises a program provided by a domestic or foreign OEM, such as Ford, General Motors, Toyota or the like, with the diagnostic application scanning tool **28** configured to enable the reading and reporting of fault codes in the electronic system of the vehicle such as may be located in ECUs of the vehicle. As noted, in practice, memory **50** may include multiple diagnostic application programs, each for use with various makes and/or models of vehicles to enable diagnosing and programming of ECUs via vehicle interface module **27**, including depending on the particular vehicle systems/ECUs on the vehicle installed by the OEM based on the vehicle purchaser's selection of vehicle options. Alternatively and/or additionally, memory **50** may include diagnostic application programs that may be used with multiple variations of vehicles. Vehicle interface module **27** and computer **30** are thus cooperatively used for querying/scanning and diagnosing ECUs of vehicle **22**, including for accessing error codes

generated by the ECUs for assessing and diagnosing operational and performance related aspects of the vehicle **22**.

(16) As illustrated in FIG. **1**, diagnostic tool **28** may receive various inputs **21** that are used, including by system list program **45**, to generate the checklist **26**. In the illustrated embodiment this may include vehicle detail inputs **25**, vehicle safety system configuration inputs **19**, restraint deployment status inputs **31**, and vehicle damage inputs **35**.

(17) Vehicle detail inputs **25** may comprise various information, including the make, model and/or year of the vehicle **22**. The vehicle detail inputs **25** may further or alternatively include the vehicle identification number (“VIN”) of the vehicle **22**. The vehicle detail inputs **25** may be manually entered by mechanic into diagnostic tool **28**, such as via touch screen **66a** or keyboard **66b**. Alternatively, vehicle detail inputs **25** may be obtained by diagnostic tool **28** by reading from electronic system **23**, where diagnostic tool **28** may query electronic system **23**, such as to obtain the VIN. System **10** may further employ a VIN database **37** via which specifics of vehicle **22** may be ascertained by diagnostic tool **28**. For example, VIN database **37** may be used to determine the make, model and/or year of vehicle **22** via the VIN.

(18) Safety system inputs **19** are provided to or are determined by diagnostic computer tool **28** to identify the safety systems equipped on the vehicle, including ADAS systems and airbag modules on vehicle **22**, such as the safety systems on the vehicle as built or equipped by the vehicle manufacturer when built, or the existing safety systems on the vehicle. It should be appreciated that different vehicles may be equipped with different ECUs when built, including that some vehicles of the same make/model may have different or additional ECUs when compared to other vehicles of the same make/model, such as based on options or equipment packages. For example, a given vehicle may be sold with or without various advanced driver-assistance systems (ADAS), such as adaptive cruise control, lane departure warning, parking assistance, blind spot detection, collision avoidance, forward collision warning, surround view, automatic parking, and other driver assistance systems or other vehicle options. Each ADAS subsystem may include its own additional ECUs, or an ECU may be shared for more than one ADAS feature, where such additional ECUs must be queried/scanned, accessed, and then assessed. In particular, such safety systems may require calibration, such as to confirm proper operation, including if such systems required repair or replacement as a result of collision damage. In accordance with aspects of the present invention, therefore, prior to generating a checklist **26** for providing a mechanic **24**, an evaluation may be performed to determine the vehicle systems present on the subject vehicle whereby the checklist **26** may be generated to only include systems that are on the specific vehicle **22** at issue, including based on the particular systems/ECUs present on the vehicle and/or any particular damage that occurred to vehicle **22**, such as from a collision. The configuration of vehicle safety systems of a vehicle may be ascertained via an onboard or offboard determination.

(19) Safety system inputs **19** may be obtained in a number of ways, including using either one or more onboard databases or offboard databases. In one onboard configuration, diagnostic tool **28** includes a diagnostic evaluation program application **51** (FIG. **3**) for querying/scanning the vehicle **22** to read vehicle data information regarding the particular ECUs that are present on the vehicle, with the vehicle data information being analyzed or parsed to determine the particular vehicle systems provided on the vehicle based on the particular ECUs that are detected. In particular, system **10** determines whether vehicle **22** includes particular safety systems, such as any ADAS systems based on the particular ECUs that are detected. Accordingly, prior to generating a checklist **26** for a given vehicle, a diagnostic evaluation program **51** may initially be run to read or obtain vehicle data information related to the equipped vehicle systems on the vehicle. In one embodiment, vehicle diagnostic tool **28** includes an onboard memory or database **50b** that includes known ECUs that may be present on a vehicle, including ECUs for ADAS systems on a vehicle. Diagnostic evaluation program **51** is run to determine the ECUs that are present on the vehicle **22**, where the detected ECUs may be compared to the known ECUs identified in database **50b** that may

be present, whereby the particular ECUs present on vehicle **22** may be identified. In one embodiment, information contained within database **50b** includes information regarding addresses within electrical system **23** for the potential ECUs of vehicle **22**, where the addresses may be specified, for example, by a bus address, such as a unique memory address or CAN address for the various addresses of the ECUs. For example, database **50b** may include information regarding each possible ECU that was available for a given make and model of vehicle, such as by year, including for all potential options, including ADAS systems, including the specific ECU for each possible module and the address within electrical system **23** that such ECU is located. Diagnostic evaluation program **51** may operate to query each address of electrical system **23** at which an ECU may be present, such as by sending an inquiry signal to such addresses, where system **10** is able to confirm the presence of a given ECU for such addresses if a response signal is received from an ECU at the given address. Moreover, system **10** is able to identify the particular ECU, including whether the ECU is an ADAS ECU based on the address, which identification information may be stored in database **50b** along with the given address. For example, each potential ECU of a given make and model of vehicle, including by year or years, may be mapped out such that database **50b** includes the address of each ECU, and may also include the specific operation, purpose or function of the ECU, such as by part number, name, or the like. System **10** may serially query addresses of electrical system **23** to determine the presence of particular ECUs, including ADAS ECUs, or may alternatively broadcast inquiry signals to multiple or all potential addresses in parallel. Still further, rather than query each ECU within the electronic system **23**, system **10** may be configured to query only for ADAS ECUs. In such a configuration database **50b** may only contain ECU addresses for ADAS systems. Still further, as discussed in more detail below, a remote or offboard database may be employed for determining the safety systems present on vehicle **22**, where for example, a database such as database **50b** may be disposed at a remote computer, such as server **64**.

(20) In an alternative configuration or operation, system **10** may obtain safety system inputs **19** by determining the ADAS systems present on vehicle **22** by way of build data from the vehicle database **50b** that comprises a VIN database. For example, the diagnostic evaluation program **51** may obtain the vehicle VIN upon connection of system **10** with vehicle **22**. As a first step, system **10** via the vehicle diagnostic tool **28** may initially acquire a vehicle identification number (VIN) associated with a vehicle under test. In an aspect of the present invention, the diagnostic tool **28** is operable to read the VIN from the vehicle **22** via its connection through the OBD2 diagnostic port **32**. The vehicle VIN may then be used to determine the vehicle systems equipped on the vehicle, such as via VIN database **50b**, where for example, computer **30** may include database **50b** that is operatively used to determine the vehicle systems on the vehicle based on the determined VIN. This may include an algorithmic lookup table based on the identified VIN, such as for example where certain alphanumeric characters of the VIN identify the presence or absence of particular vehicle systems present on the vehicle, such as ADAS systems or other vehicle or safety systems. Alternatively, the VIN for the vehicle **22** may be acquired through alternative means and directly input by the operator **24**, such as by being visually examined and input via interface **66**, such as by way of a keyboard or touch screen. Moreover, rather than a lookup VIN database, the system **20** may operatively algorithmically analyze selected alphanumeric characters, such as by position number in the VIN, with the system **20** recognizing based on the particular character and location the presence or absence of particular vehicle systems present on the vehicle, such as being preprogrammed. This may include, for example, an operator initially entering a make and model of a vehicle via interface **66** whereby the system **20** is preconfigured to read particular characters in particular locations of the VIN in order to determine the presence or absence of particular vehicle systems present on the vehicle. Again, as noted below, rather than being configured as an onboard database, database **50b** may alternatively comprise an offboard database, such as residing at remote server **64**.

(21) An exemplary embodiment of a database **50b** configured as a VIN database is illustrated in

FIG. 3A, such as for one particular make and model of vehicle. As there shown, database **50b** includes a listing of VIN data **80a**, **80b**, **80c**, where VIN data may comprise complete VIN numbers, ranges of VIN numbers, or selected portions or ranges of VIN numbers for a given make and model vehicle, with the VIN data **80a**, **80b**, **80c** being correlated with vehicle systems **81a**, **81b**, **81c**, where the vehicle systems **81a**, **81b**, **81c** in the illustrated embodiment are ADAS systems such as a forward facing lane keeping assist system, an adaptive cruise control system, and a collision avoidance system. It should be appreciated that although database **50b** is illustrated in FIG. 3A to include three separate VIN data designations and three separate vehicle systems, that in practice the VIN database may include numerous listings of VIN data as well as numerous vehicle systems. The vehicle database **50b** may further comprise a VIN/ECU part number database that includes a listing of ECU part numbers associated with a given VIN, such as based on build data, including such as may be provide by an OEM. Accordingly, generation of checklist **26** may be based in part upon a review of the VIN database and/or a review of the ECU part numbers associated with the current vehicle's VIN.

(22) Still further, safety system inputs **19** comprising the ADAS systems present on a vehicle may be obtained by way of vehicle database **50b** based on the year, make and model of a vehicle **22**. In such a configuration the vehicle database **50b** may identify the possible ADAS systems on vehicle **22** based on the year, make and model of vehicle **22**. As noted above, this may be acquired via diagnostic tool **28** reading the electronic system **23**, or may be entered by the mechanic **24**, such as via screen **66a**.

(23) Although database **50b** is shown as being within memory **50** of computer device **30**, it should be appreciated that database **50b** may alternatively be remotely located, such as at server **64** and accessible via internet connection **70**.

(24) It should be further appreciated that in the step of determining the presence of the particular ECUs on the vehicle **22**, that the absence of a response signal from an ECU at a given address, such as an ADAS ECU, may mean either that the module and ECU was not present on the vehicle, i.e. it was not originally equipped with the ADAS module and associated ECUs, or that the module having such ECU has been damaged. For example, in the case of a vehicle that has been damaged, such as by a collision, it is possible that the vehicle may have been equipped with particular vehicle systems or modules having ECUs, such as ADAS ECUs, but that due to the damage to the vehicle the diagnostic evaluation program **51** may not be able to detect all of the ECUs due to the vehicle damage. For example, a vehicle may be equipped with certain exterior cameras, radar or ultrasonic sensors, or other ADAS equipment. If such components are damaged in a collision, and or associated controllers are damaged in a collision, then diagnostic evaluation program **51** may inadvertently infer that such systems were not present on vehicle when instead they are not responding or detectable due to the damage. As such, in accordance with a further aspect of the present invention, inputs **21** of system **10** may additionally include inputs **35** related to damaged areas of vehicle **22** whereby system **10** takes into consideration the damage and the potentially effected vehicle systems and associated ECUs, including ADAS systems and associated ADAS ECUs. With reference to FIG. 1, damage information inputs are noted at **35** in FIG. 1.

(25) Damage information inputs **35** may be provided to system **10** via one or more various sources. In one configuration, one or more images representative of the particular vehicle **22** being diagnosed may be shown on screen **66a** of diagnostic tool **28** where, for example, based on a detected VIN, diagnostic tool **28** may display an image representative of the make, model and year of vehicle **22**. The mechanic **24** may then be prompted to interact with the displayed image, such as with screen **66a** being configured as a touchscreen, to designate or highlight on the displayed image the location or locations corresponding to any actual damage on vehicle **22**. Alternatively, in another configuration system **10** may obtain digital images or photographs of the actual vehicle **22** requiring repair and, via image recognition software, determine the location or locations of damage on vehicle **22**. Such digital images may be provided to system **10** from a separate camera, or for

example, diagnostic tool **28** may include an integrated camera or imager, such as a CMOS imager, with which to take digital images of vehicle **22**, with image recognition software residing in computer module **30**, such as in memory **50**. Still further, in yet another configuration system **10** may interface with a collision estimating software program, such as via an API exchange, whereby system **10** may receive a predetermined evaluation of any damage to vehicle **22**, including to specific vehicle systems whereby the damaged ADAS modules are determined. For example, information regarding physical damage to a vehicle via is illustrated as being provided to diagnostic tool **28** from a collision estimating software residing on a separate computer **65** in FIG.

1. It should be appreciated that the collision estimating software may alternatively reside on diagnostic tool **28** itself, or be accessed directly via diagnostic tool **28**.

(26) As also understood from FIG. **1**, an additional input **21** comprises restraint deployment inputs **31**. Restraint deployment inputs **31** include an identification of whether or not one or more of the vehicle airbags have been deployed, which may be entered by a mechanic **24** into diagnostic tool **28** by way of a visual inspection of vehicle **22**, with diagnostic tool **28** providing a prompt to direct and allow entry of a response by mechanic **24**. Alternatively, diagnostic tool **28** may be used to perform or run a pre-scan diagnostic application program, which may be a limited pre-scan, that determines if any trouble codes for the vehicle airbags are detected indicating that airbags on vehicle **22** have been deployed. The restraint deployment inputs **31** may additionally include an identification of whether or not one or more of a seat belt tensioner or pretensioner has been activated or deployed. For example, some vehicles include electronic seat belt pretensioners that utilize an explosive charge and include an ECU for providing tension to a seatbelt in the event a collision is detected. Accordingly, in like manner to determining deployment of airbags, deployment of seatbelt pretensioner restraints may be determined by visual inspection by mechanic **24** and entry into diagnostic tool **28**, or by use of diagnostic tool **28** to perform a pre-scan to detect trouble codes indicating deployment of the seatbelt pretensioner system.

(27) It should be appreciated that the ADAS system identification herein for purposes of the present invention may include or encompass identification of airbag modules requiring programming, and that calibration for purposes of the present invention may include or encompass the programming of the airbag modules.

(28) Upon determining the safety systems and damage present on vehicle **22**, system list program **45** is operable to generate checklist **26** that in one embodiment is presented to mechanic **24** on screen **66** of computer **30**. As understood from FIG. **4**, in the illustrated embodiment checklist **26** includes buttons or display fields **26a**, **26b**, **26c** listing various vehicle systems **81a**, **81b**, **81c**, which comprise ADAS safety systems that have been identified via system list program **45** as being present on vehicle and associated with damage to vehicle **22**, and which accordingly may require calibration operations to be performed by the mechanic **24**, where the systems **81a**, **81b**, **81c** would include various ADAS ECUs **41**. That is, the vehicle systems **81a**, **81b**, **81c** are or include components that are disposed in locations in which damage has been identified on vehicle **22**, where in the illustrated embodiment the ADAS systems comprise a forward facing lane keeping assist system **81a**, an adaptive cruise control system **81b**, and a collision avoidance system **81c**. It should be appreciated that each of these ADAS systems may comprise numerous components, including for example various cameras, radar modules and associated ECUs, and that various of such components may be utilized on more than one given ADAS system. Calibration operations for the ADAS systems **81a**, **81b**, **81c** may be performed by the mechanic **24** such as part of the repair and/or replacement of components of such ADAS systems and/or to confirm operation of the systems.

(29) Checklist **26** in the illustrated embodiment additionally includes buttons or display fields **126a**, **126b**, **126c** associated with each of the ADAS systems **81a**, **81b**, **81c** that are operable to provide an indicator to the mechanic as to whether or not each of the systems **81a**, **81b**, **81c** have been calibrated, where fields **126a**, **126b**, **126c** may be toggled from not calibrated to completed

upon performance of the appropriate calibration or calibration procedure for the given ADAS system. Still further, checklist **26** may include additional buttons or fields **226a**, **226b**, **226c** associated with each of the ADAS systems **81a**, **81b**, **81c** displayed on the checklist **26** that indicate whether or not the diagnostic computer tool **28** supports calibration of the associated ADAS system **81a**, **81b**, **81c**. Moreover, if diagnostic computer tool **28** supports calibration of the given ADAS system, mechanic **24** may click the associated button **226a**, **226b**, **226c** for that supported ADAS system and launch directly to the calibration function or support function for the given ADAS system. This beneficially avoids the mechanic **24** from having to navigate through to alternative programs, including avoiding having to enter information to access the appropriate calibration materials for a given ADAS system.

(30) For example, with reference to FIGS. **2** and **3C**, diagnostic computer tool **28** may include stored electronic calibration data **50c** retained within memory **50** for support of calibration of one or more ADAS systems, such as systems **81a**, **81b**, **81c**. It should be appreciated that the calibration operation for a given ADAS system **81a**, **81b**, **81c** may be dependent upon the given system as well as the year, make and model of vehicle **22** at issue. Accordingly, memory **50** may include distinct electronically stored calibration data or information, such as shown at A, B, C, D in FIG. **3C**, for use in calibrating various different ADAS systems, where the stored calibration data may comprise one or more various calibration related instruction files or programs, such as illustrated as A1, A2, B1, B2, C1, C2, D1 and D2. Although shown in the illustrated embodiment as being located in memory **50**, it should be appreciated that some or all of the calibration data **50c** may be remotely located, such as in server **64**, and accessible via Internet connection **70**.

(31) In operation, the electronic calibration data **50c** provides instruction files to a mechanic **24** comprising human readable information that may be displayed on screen **66a** or printed for calibration of the given ADAS system and/or provides executable calibration programs that interact with ECUs of the ADAS system for conducting, completing and confirming calibration of the ADAS system. With respect to the illustrated embodiment of FIG. **4**, for example, fields **226a**, **226b**, **226c** indicate that diagnostic computer tool **28** supports calibration operations for the given ADAS systems **81a**, **81b**, **81c** indicated at **26a**, **26b**, **26c** as requiring calibration. Mechanic **24** may launch particular calibration support data **50c** by selecting one of the associated fields **226a**, **226b**, **226c**, where this may be done in any of various manners, such as via depressing a touchscreen or selecting via a cursor such as with a mouse. Upon selecting one of the associated supported fields **226a**, **226b**, **226c** to directly launch an associated calibration instruction file, screen **66a** may provide instructions to mechanic **24** regarding the calibration procedure for the given ADAS system. This may include, for example, providing a list of step-by-step instructions for the mechanic **24** to undertake to calibrate the given ADAS system, this includes identifying the particular equipment needed, such as targets for sensors of the ADAS system, as well as the location in which to position the targets and/or vehicle relative to each other. Still further, electronic calibration data **50c** may include calibration programs, such as OEM and/or aftermarket ADAS calibration programs, for running on ADAS ECUs **41** as part of the calibration process. For example, the ADAS calibration program may be run via vehicle interface **27** on electronic system **23**. Diagnostic computer tool **28** may also require or prompt mechanic **24** to verify completion of each given step, such as by way of entry via screen **66a** and/or keyboard **66b**. Completion of the various steps may be confirmed in various ways, including for example where completion of some or all of the steps is automatically monitored by diagnostic computer tool **28** and/or by confirmation entries by mechanic **24** via diagnostic computer tool **28**. The ADAS calibration routine may additionally include running of calibration programs, such as program A1 for a given ADAS system. The calibration program may interface with an ECU in any of various known manners, including sending a command to begin a learn function with diagnostic computer tool **28** waiting for a response from the ECU indicating completion, or the calibration program may transmit persistent messages to an ECU for calibration, or the calibration program in diagnostic

computer tool **28** may trigger or activate an internal OEM program of the ECU used for calibration. (32) Upon completing all of the steps and running of the associated calibration programs, diagnostic computer tool **28** may be returned to display checklist **26**, which is then updated to reflect that a successful calibration of the ADAS system has been performed. For example, the mechanic **24** may navigate to the checklist **26** screen via computer **30** and manually toggle the associated field **126a**, **126b** or **126c** of checklist **26** to indicate that the calibration has been performed. Alternatively, upon successfully calibrating the given ADAS system, system **10** may automatically return to the screen **66a** display of the checklist **26** and automatically toggle the associated field **126a**, **126b**, or **126c** upon completion of the calibration event. System **10** may then additionally record the successful calibration event, or indication thereof, to a work file **83** (FIG. 1), where work file **83** may be retained in memory on computer **30** and/or server **64**.

(33) System **10**, such as via diagnostic computer tool **28**, may additionally transmit a report comprising work file **83** or based on work file **83**, such as to remote computer device **64** or another remote computer. For example, such a report may be transmitted to a shop owner, insurance estimator, insurance company, the vehicle owner, or the like. The report may include indication of successful completion of the calibration of all ADAS systems identified in the checklist **26** as requiring calibration. The report may additionally include or cause an invoice to be generated for transmittal, as well as generate or cause a revenue sharing portion to be provided to the shop using system **10**.

(34) It should be understood that diagnostic computer tool **28** may not include all necessary calibration files **50c** for all ADAS systems, such as for all makes, models and years of vehicles. In which case, checklist **26** will so indicate. Such unsupported ADAS systems may then require calibration via alternative means, such as directly through an OEM procedure that may be provided at an authorized dealer, or using specific equipment and systems, including calibration files, provided by a given OEM.

(35) Still further, although diagnostic computer system **28** is shown and described in the illustrated embodiment above as including calibration files **50c** retained within memory **50** of computer **30**, it should be appreciated that some or all of the calibration function operations may be accomplished via a remote computer system **64**. For example, calibration files **50c** may be retained remotely and/or some aspects of the calibration function may be divided between a remote computer system **64** and the local computer **30**. Still further, system list program **45** may communicate with remote server **64** to confirm that a calibration has been completed, such as for example if calibration files **50c** are retained on server **64**, including for example calibration programs. System **10** may additionally enable a mechanic **24** to obtain remote assistance, such as from a remote technician at computer system **64** via Internet connection **70**.

(36) FIG. 5 illustrates the general process of use of diagnostic system **10** for generating and using checklist **26**. Step **100** is identification of ADAS systems present on a vehicle and step **102** is identification of damage areas on the vehicle, where the identification of the ADAS systems and damage areas may be performed as detailed above. The checklist **26** is then generated at step **104**, such as via system list program **45**. Subsequently, as illustrated at step **106**, the checklist **26** is displayed to a mechanic, such as on a screen **66a** of computer **30**, along with displaying which of the ADAS systems listed on the checklist **26** are supported with calibration functions by the diagnostic computer tool **28**. As understood from step **108**, mechanic **24** may launch a calibration operation via checklist **26** and diagnostic computer tool **28**. At step **110**, upon successful completion of the calibration of an ADAS system listed on checklist **26**, the status identification shown on checklist **26** may be toggled to complete. This step **110** may be done for each of the ADAS systems listed on checklist **26** as requiring calibration. Moreover, the identification of having completed calibration may be automatically generated by way of system list program **45** communicating with calibration files **50c**, including if calibration files **50c** are maintained on computer tool **28** and/or if maintained on remote server **64**. And as illustrated at step **112**, records

of the completed calibrations may then be saved in a work file **83** regarding the repair work for the vehicle **22**. Upon completion of the calibration operations, the mechanic **24** may then run a post scan using one of the diagnostic scanning application programs **53a**, **53b** for confirmation of completion of repairs to vehicle **22**. Moreover, vehicle diagnostic computer tool **28** may additionally be configured to prevent performance of a post scan until confirmation has been entered that all required or suggested calibrations have been performed. As noted above, such confirmation of completion of a calibration may be by an operator toggling the indication on screen **66a**, or may be automated via diagnostic computer tool **28** after successful calibration via program files **50c**.

(37) With further regard to determination of the ADAS systems present on vehicle **10** at step **100**, system **10** may additionally employ or use a vehicle ADAS database **49**, where database **49** includes information regarding all possible ADAS modules that may have been available for a given make, model and year of vehicle. For example, as noted above, diagnostic tool **28** may query electronic system **23** to obtain the VIN of vehicle **22**, or the mechanic **24** may enter the VIN into diagnostic tool **28**, such as a mechanic input **47**. Alternatively, the mechanic may input the make, model and year of the vehicle **22** as a manual input. Based on the determined and decoded VIN and/or the make, model and year of the vehicle **22**, database **49** is operable to determine what possible ADAS modules or systems may have been available for the vehicle **22**.

(38) In one embodiment, VIN database **37** and/or ADAS database **49** are located remotely from the vehicle repair facility and accessed by diagnostic tool **28**, such as being retained within remote server **64** and accessed via a wireless Internet connection **70**. Alternatively, databases **37** and **49** may be combined with database **50b**, either remotely or locally. Likewise, system list program **45**, diagnostic evaluation program **51**, and/or diagnostic scan programs **53a**, **53b** may also be located remotely and be accessed by diagnostic computer tool **28**. Alternatively, all or any of system list program **45**, diagnostic evaluation program **51**, and/or diagnostic scan programs **53a**, **53b** are contained within memory **50** of diagnostic tool **28**, as discussed above. Still further, it should be additionally appreciated that programs **45**, **50c**, **51**, **53a**, **53b** may each comprise multiple subparts or routines that operate together. It should be further appreciated that inputs **21**, such as vehicle safety system inputs **19**, restraint deployment inputs **31** and/or damage inputs **35** may be variables that are used by or entered into system list program **45**.

(39) In a further particular alternative embodiment, system **10** may query the electronic system **23** of the vehicle **22** to obtain the part numbers of all ECUs resident on vehicle **22**, where the ECU part numbers are readable via interface module **27** and the connection with OBD diagnostic port **32**.

(40) An exemplary embodiment of a database **50b** configured as an ECU database is illustrated in FIG. **3B**. As there shown, database **50b** includes a listing of ECU part numbers **82a**, **82b**, **82c**, such as for one particular make and model of vehicle, with the ECU part numbers being correlated with vehicle systems **81a**, **81b**, **81c**, where the vehicle systems **81a**, **81b**, **81c** in the illustrated embodiment are ADAS systems such as a forward facing lane keeping assist system, an adaptive cruise control system, and a collision avoidance system. It should be appreciated that although database **50b** is illustrated in FIG. **3B** to include three separate ECU part numbers associated with three separate vehicle systems, in practice a vehicle may be equipped with dozens or hundreds of ECUs where by the ECU database may include numerous listings of ECU part numbers as well as numerous vehicle systems. It should be appreciated that database **50b** may include both ECU part numbers and addresses, which may be correlated or combined together.

(41) The vehicle database **50b** may further comprise a VIN/ECU part number database that includes a listing of ECU part numbers associated with a given VIN. Therefore, based upon a review of the VIN database and a review of the ECU part numbers associated with the current vehicle's VIN, a selection may be made between a third party supplied scanning program or a manufacturer supplied scanning program. For example, if the VIN database reveals that a given vehicle contains one or more ECU part numbers associated with ADAS ECUs, a decision can be

made to select the more comprehensive manufacturer supplied scanning program. In the alternative, if a review of the VIN database reveals that the vehicle is not equipped with ADAS ECUs, the faster and less costly third-party supplied scanning program may be utilized. Such a decision may be aided if the VIN entry in the VIN database for an ADAS equipped vehicle includes at least one part number of an ADAS ECU associated with the VIN. Alternatively, the VIN entry in the VIN database for an ADAS equipped vehicle may include some other indication in the VIN database that the VIN is associated with an ADAS equipped vehicle. Therefore, if a vehicle's VIN entry in the VIN database does not include an ADAS ECU part number or some other indication that the VIN is for an ADAS equipped vehicle, the third-party supplied scanning program may be selected.

(42) Still further, system **10** may be configured to accept operator input to obtain vehicle data information. For example, system **10** may prompt an operator **24** to enter vehicle make and model information, such as via interface **66**. System **10** may then also provide step-by-step inquiries to operator **24** regarding systems installed on vehicle **22**, such as whether vehicle **22** includes particular cameras at particular locations, or specifically whether vehicle **22** has particular ADAS systems. Accordingly, system **10** instead of automatically launching a diagnostic scanning program **53a**, **53b** to be run after diagnostic evaluation program **51** queries vehicle electronic system **23** to determine which vehicle systems **81a**, **81b**, **81c** are present on vehicle **22**, or prior thereto, system **10** may instead provide a recommendation of whether to run an aftermarket diagnostic scan program **53a** or an OE diagnostic scan program **53b**, in particular based on various scan inputs **21**.

(43) Although database **50b** is disclosed above as residing in diagnostic computer tool **28**, such a database may additionally or alternatively reside in the memory of a remote computer, such as a remote server **64**. If the database is located in a remote server **64**, the vehicle interface tool **28** may access the database via an Internet interface. Optionally, the database, such as a database found on the remote computer or server **64**, may be a compiled database of VINs and/or ECUs provided by a third party. Still further, the database found in the memory of the vehicle interface tool **28** may be a local database that comprises a portion of the information, where the database found on the remote computer or server **64** is remote and contains additional or alternative information. For example, the database may be progressively expanded to include those VINs and/or associated ECU part numbers for vehicles previously serviced at a particular service station or automotive repair business. Optionally, the database may be updated, such that an operator may add a new vehicle VIN and associated ECU part numbers. The database may thus be a tool for identifying which vehicles are equipped with ADAS subsystems.

(44) Diagnostic tool **28** may, either in addition to the above noted various loaded diagnostic applications or in place thereof, be used to access remotely located diagnostic applications, such as that may reside on remotely located servers **64**. This may be done, for example, to avoid the need for obtaining and locally storing and maintaining diagnostic applications on diagnostic tool **28**. In the illustrated embodiment, diagnostic tool **28** includes an input/output (I/O) interface **66** for coupling to peripheral devices, such as one or more of a monitor, keyboard, mouse, and the like. In a further illustrated embodiment, the diagnostic tool **28** is implemented as a laptop computer with integrated monitor, keyboard, and mouse.

(45) As an alternative to providing checklist **26** on screen **66** of computer **30**, diagnostic computer tool **28** may instead or in addition output the associated data of checklist **26** to a report. For example, an electronic data file may be generated by the vehicle diagnostic computer tool **28** that may be provided to another computer device and/or may be printed. The electronic data file may list the vehicle safety systems equipped on the vehicle requiring calibration. Still further, the output data file may also include calibration information for safety systems supported by the diagnostic computer tool **28**. For example, the calibration information may comprise calibration instructions that may be separately printed or displayed on another computer to provide human readable instructions for calibrating safety systems.

(46) System **10**, in addition to performing scanning operations, may also be used for reprogramming of vehicle **22**, such as reprogramming selected ECUs, including reprogramming ECUs based on a determination of particular fault codes.

(47) Although vehicle interface diagnostic tool **28** is discussed above as conforming with the SAE J2534 standard, it should be appreciated that alternatively configured vehicle diagnostic and programming tools may be employed within the scope of the present invention, including alternatively configured tools for alternative types of vehicles, such as alternative classes of vehicles. Accordingly, an interface tool may conform with the ISO 22900 standard, or RP1210 standard, or may operate under the ELM327 command protocol.

(48) As noted, the software and/or hardware of diagnostic and programming tools may be required to be updated to operate with new vehicles and/or enable programming and diagnosing of existing vehicles. In the above noted embodiments the vehicle interface diagnostic tool **28** may be periodically updated via an Internet connection, or may be returned to the supplier for updating, including with regard to hardware updates. This may be done by the supplier of the vehicle interface diagnostic tool **28** whereby the local operator **24** need not spend time attempting to maintain the equipment.

(49) In the illustrated embodiment vehicle interface diagnostic computer tool **28** is disclosed as including both the diagnostic evaluation program **51** as well as multiple diagnostic scanning programs **53a**, **53b**. It should be appreciated that alternative arrangements may be employed within the scope of the present invention. For example, program **51** and/or programs **53a**, **53b** may reside on different devices. For example, an operator **24** may first utilize one device with a diagnostic evaluation program **51** to determine which diagnostic evaluation program is appropriate for the vehicle **22**, and then select from alternative computer devices to run the determined diagnostic evaluation program. Still further, although diagnostic evaluation program **51** and diagnostic scanning programs **53a**, **53b** are illustrated and discussed herein as being separate programs, one or more of such programs may be combined together and operate as subroutines.

(50) Changes and modifications in the specifically described embodiments can be carried out without departing from the principles of the present invention which is intended to be limited only by the scope of the appended claims, as interpreted according to the principles of patent law including the doctrine of equivalents.

Claims

1. A method for identifying and presenting vehicle safety systems on a vehicle for calibration by an operator, said method comprising: providing a vehicle diagnostic computer system comprising a vehicle diagnostic computer tool, wherein the vehicle diagnostic computer tool is configured to be connected to a diagnostic port of a vehicle to be in communication with an electronic system of the vehicle; providing a system list program, wherein said system list program is operable to generate a list of vehicle safety systems equipped on the vehicle requiring calibration based on inputs to said system list program; determining vehicle safety systems present on the vehicle via the vehicle diagnostic computer tool; determining areas on the vehicle in which there is damage, wherein said areas comprise damage areas, and providing vehicle damage information inputs to the system list program based on the damage areas; generating a list of vehicle safety systems equipped on the vehicle requiring calibration via said system list program based on the determined vehicle safety systems present on the vehicle and the vehicle damage information inputs, wherein the list of vehicle safety systems equipped on the vehicle requiring calibration corresponds to vehicle safety systems present on the vehicle that are disposed in damage areas; and providing to a user via the vehicle diagnostic computer tool the list of vehicle safety systems equipped on the vehicle requiring calibration; wherein the vehicle diagnostic computer tool includes a display screen, and wherein the method further comprises displaying on the screen of the vehicle diagnostic computer

tool the list of vehicle safety systems equipped on the vehicle requiring calibration; wherein the list of vehicle safety systems requiring calibration comprise Advanced Driver Assistance Systems (ADAS), and wherein said vehicle diagnostic computer system further comprises electronic calibration data configured to provide calibration support for one or more of the listed vehicle safety systems requiring calibration; and wherein the list of vehicle safety systems requiring calibration additionally includes an indication of which of the vehicle safety systems requiring calibration are supported by said vehicle diagnostic computer system via the electronic calibration data, and wherein said displaying on the screen of the vehicle diagnostic computer tool the list of vehicle safety systems equipped on the vehicle requiring calibration further comprises displaying an indication of which of the vehicle safety systems requiring calibration are supported by said vehicle diagnostic computer system, and wherein said method further comprises; accessing, by a user of the vehicle diagnostic computer tool, electronic calibration data for at least one vehicle safety system requiring calibration that is supported by said vehicle diagnostic computer system, wherein said accessing by the user comprises launching via the vehicle diagnostic computer tool electronic calibration data for the at least one vehicle safety system; and calibrating the at least one vehicle safety system based on the electronic calibration data, wherein said calibrating the at least one vehicle safety system comprises using by the user the vehicle diagnostic computer tool and electronic calibration data associated with the at least one vehicle safety system to calibrate the at least one vehicle safety system for repairing at least one of the damage areas of the vehicle.

2. The method of claim 1, wherein said vehicle diagnostic computer tool includes said system list program.

3. The method of claim 1, further comprising reading vehicle data information of the vehicle with the vehicle diagnostic computer tool, and wherein said determining vehicle safety systems present on the vehicle via the vehicle diagnostic computer tool comprises using the vehicle data information to determine the vehicle safety systems present on the vehicle.

4. The method of claim 3, wherein reading vehicle data information of the vehicle with the vehicle diagnostic computer tool comprises reading a vehicle identification number (VIN) of the vehicle.

5. The method of claim 4, wherein said determining vehicle safety systems present on the vehicle via the vehicle diagnostic computer tool comprises using a database to identify vehicle systems present on the vehicle.

6. The method of claim 1, wherein the vehicle damage information inputs comprises at least one of; one or more digital images of the vehicle, and wherein image recognition software is used on the one or more digital images to determine damage to the vehicle; data from a collision estimating software; or data input by a user of the vehicle diagnostic tool.

7. The method of claim 1, wherein said displaying on the screen of the vehicle diagnostic computer tool the list of vehicle safety systems equipped on the vehicle requiring calibration further comprises displaying on the screen of the vehicle diagnostic computer tool an indication of completion of calibration of vehicle safety systems upon completion of calibration thereof.

8. The method of claim 1, wherein the vehicle diagnostic computer tool is operatively connected with a remote computer system, and wherein the system list program interfaces with the remote computer system, with the remote computer system configured to confirm that all of the listed vehicle safety systems equipped on the vehicle requiring calibration have been calibrated.

9. The method of claim 1, further comprising running a vehicle diagnostic scan using said vehicle diagnostic computer tool, and wherein prior to running the vehicle diagnostic scan said vehicle diagnostic computer tool requires confirmation that all of the listed vehicle safety systems equipped on the vehicle requiring calibration have been calibrated.

10. The method of claim 9, wherein the vehicle diagnostic scan comprises a post scan after performing repairs on the vehicle.

11. A vehicle diagnostic system comprising: a vehicle diagnostic computer tool, wherein the vehicle diagnostic computer tool includes a display screen, and wherein the vehicle diagnostic

computer tool is configured to be connected to a diagnostic port of a vehicle to be in communication with an electronic system of the vehicle; a system list program; electronic calibration data configured to provide calibration support for one or more vehicle safety systems; wherein said vehicle diagnostic tool is configured to determine vehicle safety systems present on the vehicle, and wherein said vehicle diagnostic tool is configured to receive vehicle damage information inputs corresponding to areas on the vehicle in which there is damage, wherein said areas comprise damage areas; wherein said system list program is configured to generate a list of vehicle safety systems equipped on the vehicle requiring calibration based on the determined vehicle safety systems present on the vehicle and the vehicle damage information inputs, wherein the list of vehicle safety systems equipped on the vehicle requiring calibration corresponds to vehicle safety systems present on the vehicle that are disposed in damage areas; wherein said vehicle diagnostic computer tool is configured to display on the screen of the vehicle diagnostic computer tool the list of vehicle safety systems equipped on the vehicle requiring calibration and an indication of which of the vehicle safety systems requiring calibration are supported for calibration via the electronic calibration data; and wherein said vehicle diagnostic computer tool is configured to enable a user to launch via the vehicle diagnostic computer tool electronic calibration data for each of the vehicle safety systems equipped on the vehicle requiring calibration and that are supported for calibration for repairing the damage areas of the supported vehicle safety systems.

12. The vehicle diagnostic system of claim 11, wherein the vehicle diagnostic computer tool comprises the system list program.

13. The vehicle diagnostic system of claim 11, wherein the vehicle diagnostic computer tool is configured to read vehicle data information of the vehicle to determine the vehicle safety systems present on the vehicle.

14. The vehicle diagnostic system of claim 13, wherein the vehicle data information of the vehicle comprises a vehicle identification number (VIN) of the vehicle, and wherein the vehicle diagnostic computer tool is configured to interface with a database to determine the vehicle safety systems present on the vehicle based on the VIN.

15. The vehicle diagnostic system of claim 11, wherein the vehicle damage information inputs comprise at least one of; one or more digital images of the vehicle, and wherein image recognition software is used on the one or more digital images to determine damage to the vehicle; data from a collision estimating software; or data input by a user of the vehicle diagnostic tool.

16. The vehicle diagnostic system of claim 11, wherein the vehicle diagnostic computer tool is configured to allow a user to access the electronic calibration data with the vehicle diagnostic computer tool for calibration of vehicle safety systems requiring calibration that are supported by said vehicle diagnostic system.

17. The vehicle diagnostic system of claim 11, wherein said system further comprises diagnostic scan application programs, and wherein the vehicle diagnostic computer tool is operable to run a selected one of the diagnostic scan application programs to diagnose the vehicle, and wherein the vehicle diagnostic computer tool is configured to require confirmation that all of the listed vehicle safety systems equipped on the vehicle requiring calibration have been calibrated prior to the vehicle diagnostic computer tool running the selected one of the diagnostic scan application programs.

18. The vehicle diagnostic system of claim 11, wherein said vehicle diagnostic computer tool is configured to display on the screen of the vehicle diagnostic computer tool an indicator display field associated with each listed vehicle safety system requiring calibration, wherein each indicator display field provides an indication whether the associated vehicle safety system has been calibrated.

19. The vehicle diagnostic system of claim 11, wherein said vehicle diagnostic computer tool is configured to display on the screen of the vehicle diagnostic computer tool a support display field associated with each vehicle safety system requiring calibration that is supported by said vehicle

diagnostic computer system via the electronic calibration data, and wherein the support display field is selectively activatable by a user of the diagnostic computer tool, and wherein upon activation of a given support display field electronic calibration data comprising calibration instructions for the associated vehicle safety system are provided on the display screen of the diagnostic computer tool.

20. The method of claim 1, further comprising displaying on the display screen of the diagnostic computer tool an indicator display field associated with each listed vehicle safety system requiring calibration, wherein each indicator display field provides an indication whether the associated vehicle safety system has been calibrated.

21. The method of claim 1, further comprising displaying on the display screen of the diagnostic computer tool a support display field associated with each vehicle safety system requiring calibration that is supported by said vehicle diagnostic computer system via the electronic calibration data.

22. The method of claim 21, wherein the support display field is selectively activatable by a user of the diagnostic computer tool, and wherein upon activation of a given support display field electronic calibration data comprising calibration instructions for the associated vehicle safety system are provided on the display screen of the diagnostic computer tool.
